

Bhavyasri Sanisetty

Backend Software Engineer | Python | Machine Learning

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Published machine learning research in Nature Scientific Reports (2025) and developed secure authentication systems and API-based backend services.

Professional Summary

Backend-focused software engineer building REST APIs, authentication systems, and database-driven applications using Python. Implemented multi-factor authentication and integrated machine learning models into real-world workflows.

Machine Learning Engineering Experience

Hybrid Transformer-ANN-Fuzzy Logic Model for Energy Load Forecasting — 2025

Scientific Reports (Nature Portfolio), 2025

- Designed a hybrid deep-learning pipeline combining fuzzy logic preprocessing, artificial neural networks, and transformer attention for short-term electricity load forecasting.
- Performed preprocessing and feature engineering on real-world time-series energy datasets.
- Modeled nonlinear consumption patterns and long-range temporal dependencies.
- Evaluated performance using MAE and RMSE metrics.
- Achieved approximately 15–20% lower prediction error compared to traditional and standalone ML models.
- Implemented training and evaluation pipeline using Python and PyTorch.
- Validated model robustness across seasonal variations and peak-load demand periods.

Technical Experience

Salesforce Platform Configuration

May–Jun 2024

Configured objects, validation rules, and workflow automation within the Salesforce environment. Simulated CRM business processes and UI workflows through hands-on development tasks in a guided enterprise platform setting.

Certifications

Microsoft Learn: Azure Developer, DevOps Solutions **ServiceNow:** Micro Certification **HackerRank:** Problem Solving, SQL, Python

Skills

Languages: Python, JavaScript, Java **Backend:** FastAPI, Flask, REST APIs, JWT **Frontend:** React, HTML, CSS
Database: SQLite, MongoDB **Machine Learning:** Pandas, NumPy, Scikit-learn **Tools:** Git, GitHub, Postman, Linux

Projects

Multi-Factor Authentication System — [Link](#)

- Implemented a secure authentication backend using password verification and OTP validation
- Generated and verified one-time passwords before granting access to protected routes
- Issued authentication tokens and enforced authorization checks through middleware
- Enforced access control on protected endpoints using middleware-based authorization checks
- Demonstrates separation of authentication and authorization and secure session handling

Smart Farming ML Backend Service — [Link](#)

- Developed a Flask REST API to predict suitable crops using environmental and soil parameters
- Built machine-learning inference pipeline (preprocessing → prediction → response)
- Compared multiple ML models and selected best performing configuration
- Returned structured JSON responses and logged requests for monitoring

Prompt-to-UI Generator — [Link](#)

- Built FastAPI backend that converts natural-language prompts into structured component definitions.
- Returned JSON responses consumed by a React frontend to dynamically render UI components.
- Implemented asynchronous client-server communication and request validation.

Language Translation Client — [Link](#)

- Integrated external translation APIs for real-time text and voice translation
- Processed JSON responses and handled invalid input and network failures
- Demonstrates API integration and frontend-backend communication

Academic Success Prediction (Machine Learning) — [Link](#)

- Built a machine learning model to predict student performance and dropout risk using academic and demographic data
- Performed data cleaning, feature engineering, and exploratory data analysis to identify learning patterns
- Trained and evaluated classification models to identify at-risk students
- Generated alerts and data-driven recommendations to support early intervention

Education

B.Tech Computer Science & Engineering — SRM University — 2025